

Theory Question Bank (60+)

A) Core HCI + Interaction Design Foundations

1) What is Interaction Design (Sharp, Rogers & Preece definition)?

Model answer: Interaction design is the **design of interactive products** that support how people **communicate and interact** in their **everyday and working lives**. It focuses on shaping the user's interaction with technology, not just building functionality.

2) What is Winograd's definition of Interaction Design?

Model answer: Winograd describes interaction design as the **design of spaces for human communication and interaction**. The emphasis is on shaping environments (digital/physical) that enable meaningful human actions and communication.

3) Differentiate usability goals vs user experience (UX) goals.

Model answer: **Usability goals** are more **objective and measurable** (e.g., effectiveness, efficiency, safety). **UX goals** are more **subjective and emotional**, focusing on how the product feels (e.g., enjoyable, frustrating, motivating).

4) List any four usability goals and briefly explain them.

Model answer: **Effectiveness** means users can complete tasks successfully; **efficiency** means tasks take minimal time/effort. **Safety** means preventing serious errors and supporting recovery; **easy to learn** means novices can start quickly without training.

5) Why do we involve users throughout the design process?

Model answer: User involvement reduces wrong assumptions by revealing real workflows, constraints, and pain points. It supports iteration and helps ensure the **system image** matches the user's understanding, improving usability and UX.

B) Design Rules: Principles vs Guidelines vs Standards

6) What is the difference between principles, guidelines, and standards?

Model answer: **Standards** have **high authority** and are **mandatory**, but usually **low generality** (platform/company rules). **Guidelines** are **recommendations** with lower authority; **principles** are abstract, high-level truths that guide design interpretation.

7) Why do we say heuristics are “broad brush” rules?

Model answer: Heuristics are general rules-of-thumb proven useful over time, not strict platform-specific standards. They act like a checklist to catch common usability problems early.

C) Shneiderman's Golden Rules (highly examable)

8) Explain "Strive for consistency" (Golden Rule).

Model answer: Consistency means similar actions behave the same and the interface uses stable visual patterns (layout, fonts, colors). It reduces cognitive load and makes the system predictable and easier to learn.

9) What does "cater to universal usability" mean?

Model answer: It means designing for a wide range of users, from novices to experts, often by providing **shortcuts** for frequent users while keeping the interface simple for beginners. This supports accessibility and different skill levels.

10) Explain "offer informative feedback" with an example.

Model answer: For every user action, the system should give a clear and timely response (visual/audio/haptic). Example: after clicking "Submit," show a progress state then a success confirmation.

11) What is "design dialogs to yield closure"?

Model answer: Tasks should feel like they have a clear beginning, middle, and end. Users should receive a closure message like "Payment successful" to confirm completion and reduce uncertainty.

12) Explain "permit easy reversal of actions" and why it matters.

Model answer: Users must be able to **undo/redo** actions, which reduces anxiety and encourages exploration. Easy reversal supports user control and prevents fear of making mistakes.

13) What is "support internal locus of control"?

Model answer: Users should feel they are the initiators and in control, not the system. Predictable behavior and user-driven actions reduce frustration and improve trust.

14) Explain "reduce short-term memory load" (recognition over recall).

Model answer: Working memory is limited, so interfaces should avoid forcing users to remember details across screens. Good design supports **recognition over recall** by keeping options visible and chunking information.

D) Direct Manipulation (your sir's favorite keyword trap)

15) Define direct manipulation (Shneiderman) and list its 3 properties.

Model answer: Direct manipulation is interacting with visible objects on screen using physical actions rather than complex commands. Its 3 properties are **continuous representation**, **physical actions** (click/drag), and **rapid reversible actions with immediate feedback**.

16) Why are “rapid reversible actions” linked with “undo” and user confidence?

Model answer: Rapid reversible actions mean users can quickly undo/redo without penalty, which reduces anxiety and supports exploration. This creates a strong sense of control and encourages trial-and-error learning.

17) Give two benefits and one limitation of direct manipulation.

Model answer: Benefits: novices learn quickly and experts perform fast with fewer error messages; users feel control and confidence. Limitation: not all tasks map well to objects and it can waste screen space or be slower than keyboard shortcuts.

18) Differentiate interaction type vs interface style.

Model answer: **Interaction type** describes what the user is conceptually doing (instructing, conversing, manipulating, exploring, responding). **Interface style** is the mechanism used (menus, command line, gesture, voice), so they are not the same thing.

E) Mental Models + Norman Frameworks (must-use words)

19) What is a mental model in HCI?

Model answer: A mental model is the user’s internal understanding of how a system works, used to predict outcomes and choose actions. If the model is wrong or incomplete, users make consistent “logical” errors.

20) Explain the thermostat example and what it teaches.

Model answer: Many users assume “higher setting heats faster,” using a gas-pedal model, but a thermostat is closer to an on/off switch. The lesson: design should support correct mental models through clear feedback and transparent controls.

21) Define Gulf of Execution and Gulf of Evaluation (Norman).

Model answer: **Gulf of execution** is the gap between user goals and available actions (“What can I do? How do I do it?”). **Gulf of evaluation** is the gap between system state and user understanding (“What happened? Did it work?”).

22) How do we bridge the gulfs?

Model answer: We bridge the gulf of execution by making actions obvious through visibility/signifiers, and bridge the gulf of evaluation through clear immediate feedback. This reduces cognitive effort and uncertainty.

23) Explain Norman’s three-model relationship: Designer’s Model, System Image, User’s Model.

Model answer: The **designer’s model** is how designers think the system works; the **system image** is what the user actually sees (interface, manuals, help). The goal is aligning the system image with the designer’s model so the user’s model becomes accurate.

F) Cognition: attention, memory, perception (high-yield theory)

24) Define cognition and give examples relevant to HCI.

Model answer: Cognition is the mental processing involved in gaining knowledge—thinking, remembering, learning, decision-making, reading, and reasoning. HCI design affects how easily users perceive information, focus attention, and form memories during tasks.

25) Differentiate experiential vs reflective cognition (Norman).

Model answer: **Experiential cognition** is “in the moment” acting and reacting using well-learned skills. **Reflective cognition** involves deliberate thinking, comparing, and decision-making that can generate new ideas.

26) Differentiate fast vs slow thinking (Kahneman) and why it matters in UI.

Model answer: **Fast thinking (System 1)** is automatic, intuitive, and low effort; **slow thinking (System 2)** is effortful and analytical. Interfaces should reduce unnecessary System 2 load by making choices recognizable and simple.

27) Why is “recognition over recall” a golden concept?

Model answer: Recognition is easier because users can pick from visible options rather than retrieving commands from memory. Good UI uses menus/icons and meaningful labels to minimize recall demands and reduce errors.

G) Distributed / External / Embodied Cognition

28) Define distributed cognition with an example.

Model answer: Distributed cognition says thinking is spread across people, artifacts, and the environment—not only inside one person’s head. Example: an airplane cockpit is a cognitive system involving pilot, instruments, and air-traffic control.

29) What is external cognition and cognitive offloading?

Model answer: External cognition is how we use external representations (notes, maps, diagrams) to extend thinking. Cognitive offloading uses reminders/to-do lists to reduce memory load and support decision-making.

30) What is embodied interaction?

Model answer: Embodied interaction says cognition is shaped by our physical engagement with the world; we “think with our bodies.” Touch, gestures, tangible interfaces, and VR change how users understand and act.

H) Prototyping + Fidelity (short, keyword-rich)

31) Why do we prototype before development?

Model answer: Prototyping helps test and refine ideas early when changes are cheap, and communicates design intent to stakeholders. It also exposes usability problems before coding locks the design.

32) Differentiate sketch, wireframe, and prototype.

Model answer: A **sketch** is fast ideation and disposable; a **wireframe** is the structural blueprint of layout and information architecture. A **prototype** simulates interaction and expresses design intent, sometimes across an end-to-end flow.

33) Define fidelity and name the 3 fidelity dimensions.

Model answer: Fidelity is how closely a prototype matches the final product. It varies across **visual design**, **content realism**, and **interactivity**.

34) Give two advantages and one disadvantage of low-fidelity prototyping.

Model answer: Low-fi is quick and cheap, encourages iteration, and users critique it freely because it looks unfinished. A disadvantage is low realism, so some interaction issues may not appear without a facilitator.

35) Give two advantages and one disadvantage of high-fidelity prototyping.

Model answer: High-fi supports realistic interaction and better performance metrics like time/errors, and it's persuasive for demos. A disadvantage is cost/time and the "sunk cost" effect where teams resist changes after investing effort.

I) Evaluation Methods + Testing Types (this will hit your (c) + (d))

36) Differentiate usability testing vs experiments vs in-the-wild studies.

Model answer: **Usability testing** is controlled and focuses on improving product performance.

Experiments test hypotheses and aim to discover new knowledge, while **in-the-wild** studies observe real use in natural settings over time.

37) What data do we collect in lab usability testing?

Model answer: Quantitative metrics include time on task, error counts, navigation paths, and recovery time. Qualitative data includes video, logs, and satisfaction surveys, often observed via facilitator + observation room setup.

38) What is remote testing? Name synchronous and asynchronous.

Model answer: Remote testing occurs when evaluator and participant are separated by space (and sometimes time). **Synchronous** is live (Zoom/Teams), while **asynchronous** is self-paced with screen/task recording.

39) Name the 4 usability test types and when each is used.

Model answer: **Exploratory/formative** is early to learn and shape design;

assessment/summative judges overall performance near completion. **Validation/verification** checks against usability goals, and **comparison** compares two designs or versions.

40) Why can 5–10 participants be “enough” in usability testing?

Model answer: Small samples often reveal the major recurring usability problems because many issues repeat across users. Usability testing aims to find severe problems quickly rather than statistically represent the population.

J) “Techniques for building usability in” (perfect for “which method for which user group”)

41) What is ethnographic research and why is it valuable?

Model answer: Ethnography observes users in their natural environment (“in situ”) to capture real tasks, goals, and context. It reveals implicit workflow and produces artifacts like personas, scenarios, and task descriptions.

42) What is participatory design?

Model answer: Participatory design actively involves users as co-designers rather than only test participants. It helps surface real needs and increases acceptance because users influence decisions directly.

43) What is a focus group and what is it best for?

Model answer: A focus group is a moderated discussion with multiple participants to explore attitudes, expectations, and reactions. It’s good for early idea generation and understanding perceptions, but not for measuring task performance.

44) What is card sorting (open vs closed) used for?

Model answer: Card sorting helps design **information architecture** by learning how users group and label content. **Open** sorting lets users create categories; **closed** sorting uses predefined categories to validate structure.

45) What is a walkthrough in evaluation?

Model answer: A walkthrough is a structured review where evaluators step through tasks to predict user problems and breakdowns. It’s useful when you want early feedback without running a full usability test.

K) “Technology Myopia” + Design Stages (theory-heavy)

46) Define technology myopia and why it is dangerous.

Model answer: Technology myopia is over-focusing on the technical solution while ignoring the

human and social context of use. It fails because real work includes emotions, politics, and implicit knowledge that must be supported by the design.

47) What is conceptual design?

Model answer: Conceptual design is the high-level phase where we define the system's core concepts and structure. It focuses on how information and functions are organized in a way users can understand.

L) Emotional Interaction + Affective Computing (very examable)

48) What is emotional interaction in HCI?

Model answer: Emotional interaction studies how technology makes users feel—trust, joy, anxiety, frustration—and designs for desired emotional responses. It goes beyond efficiency to include experience and engagement.

49) Explain Norman's three levels of design: visceral, behavioral, reflective.

Model answer: **Visceral** design is immediate appearance/feel (gut reaction), **behavioral** is usability and performance during use, and **reflective** is meaning, memory, and self-image. Strong products manage emotions at all three levels.

50) What is affective computing (Emotional AI)?

Model answer: Affective computing is building systems that recognize, interpret, and simulate human emotions. It uses signals like facial expressions, voice features, and biometrics to personalize experiences—raising privacy concerns.

51) What are Plutchik's 8 basic emotions?

Model answer: Plutchik lists **joy, trust, fear, surprise, sadness, disgust, anger, anticipation**. Designers use this wheel to align color palettes and mood boards with emotional intensity.

52) Define persuasive technology and nudging with examples.

Model answer: Persuasive technology designs interfaces to change behavior using psychological triggers. **Nudging** alters behavior without forbidding options, e.g., reminders to stand, default opt-in/opt-out choices, and social proof.

53) What is anthropomorphism and what is the "uncanny valley" risk?

Model answer: Anthropomorphism gives human-like qualities to systems to increase engagement and trust. But too much human-likeness can feel creepy or deceptive (uncanny valley) and creates false expectations of intelligence.

54) Why did Clippy fail compared to "Anna"?

Model answer: Clippy failed mainly because it was intrusive and interrupted tasks, creating frustration and annoyance. "Anna" was perceived as less intrusive and used lighter expressive behavior, so users tolerated it more.

55) What is zoomorphism and why can it work better than anthropomorphism?

Model answer: Zoomorphism designs systems as animals (virtual pets/robots), which lowers user expectations of intelligence. This often increases acceptance and emotional bonding without the same “creepy” human-like pressure.

M) Ubiquitous Computing + AR/VR (modern theory)

56) Define ubiquitous computing (UbiComp) and its core philosophy.

Model answer: Ubiquitous computing fills the real world with computers so they fade into the background and assist invisibly. The core idea is computation becomes implicit and context-aware rather than a “box on a desk.”

57) Differentiate VR and AR.

Model answer: **VR** recreates the real world inside a computer, immersing the user in a virtual environment. **AR** overlays digital content onto the real world, blending physical and digital context.

58) What are Weiser’s device scales (tabs/pads/boards) and why are they important?

Model answer: Weiser proposed computing scales like **inch (tabs/pads)**, **foot (tablets/notebooks)**, and **yard (boards/displays)** to match everyday objects. This matters because interaction design must fit physical context and distributed usage patterns.

N) Conceptual Knowledge: paradigm, theory, model, framework

59) Define theory, model, and framework in HCI (difference).

Model answer: A **theory** explains and predicts phenomena and identifies relevant factors. A **model** is a simplified representation used to predict or evaluate designs, while a **framework** organizes related concepts/questions to guide what to look for and how to design.

60) List roles of theory in HCI.

Model answer: Theory can be descriptive and explanatory (what/why happens), predictive (what will happen), and prescriptive/generative (how to design better). It also supports critical and conceptual thinking about technology and society.

EXTENDED THEORY QUESTION BANK (DEEP ANSWERS)

1. Cognition & Human Information Processing

Q1. What is cognition, and why is it central to Human–Computer Interaction?

Model Answer:

Cognition refers to the mental processes involved in acquiring, processing, storing, and using information, such as perception, attention, memory, learning, reasoning, and decision-making. In HCI, cognition is central because every interaction requires users to perceive interface elements, interpret meaning, remember steps, and decide actions. Poorly designed interfaces overload cognition, leading to errors and frustration. Understanding cognition allows designers to reduce cognitive load and align interfaces with human mental capabilities rather than machine logic.

Q2. Explain the difference between experiential cognition and reflective cognition.

Model Answer:

Experiential cognition refers to fast, automatic, and intuitive mental processing that happens while we are “in the moment,” such as scrolling, clicking familiar buttons, or reacting to notifications. Reflective cognition is slower and more deliberate, involving conscious thinking, evaluation, comparison, and reasoning, such as deciding which plan to buy or reflecting on past usage. In HCI, good interfaces support experiential cognition for routine tasks while enabling reflective cognition for complex decision-making. Overloading users with reflective thinking during simple tasks reduces usability.

Q3. Describe fast thinking and slow thinking and their implications for interface design.

Model Answer:

Fast thinking (System 1) is automatic, intuitive, and requires little mental effort, while slow thinking (System 2) is analytical, effortful, and conscious. Interfaces should primarily support fast thinking by using familiar conventions, clear visual cues, and recognition rather than recall. When users are forced into slow thinking for simple actions, interaction becomes tiring and error-prone. Designers must carefully decide when slow thinking is necessary, such as in critical or irreversible actions.

2. Attention, Perception & Cognitive Load

Q4. What is attention, and how does limited attention affect interface design?

Model Answer:

Attention is the cognitive process of selectively focusing on certain information while ignoring other stimuli. Human attention is limited, meaning users cannot process all interface elements simultaneously. In interface design, this requires prioritization of information, clear visual hierarchy, and reduction of unnecessary elements. Poor attention management leads to missed controls, errors, and increased mental effort, especially in high-stress or multitasking environments.

Q5. Explain the concept of information overload and its consequences in HCI.

Model Answer:

Information overload occurs when users are presented with more information than they can effectively process at one time. In HCI, this results in confusion, slower task completion, decision paralysis, and errors. Interfaces that display excessive text, controls, or options force users into constant filtering and interpretation. Good design reduces overload by progressive disclosure, grouping related items, and showing only relevant information at each stage of interaction.

Q6. Why is perception important in interaction design?

Model Answer:

Perception determines how users see, hear, and interpret interface elements such as text, icons, colors, and sounds. Designers must account for perceptual limitations, including color blindness, contrast sensitivity, and visual acuity. If users cannot perceive critical information correctly, interaction fails regardless of system functionality. Effective perceptual design ensures visibility, legibility, and distinguishability of interface components.

3. Memory & Mental Load

Q7. Explain the difference between recognition and recall, and why recognition is preferred in UI design.

Model Answer:

Recognition involves identifying information from visible options, while recall requires retrieving information from memory without cues. Recognition places far less cognitive load on users and is therefore preferred in interface design. Menus, icons, and visible options support recognition, whereas command-based interfaces rely heavily on recall. Since short-term memory is limited, designs that force recall increase errors and reduce usability.

Q8. Why is Miller's 7±2 rule often misapplied in interface design?**Model Answer:**

Miller's 7±2 rule describes the number of items people can hold in short-term memory, but it applies to chunked information, not raw interface elements. Designers often misuse it to justify arbitrary menu sizes or content limits. In reality, meaningful grouping and structure matter more than exact item counts. Over-reliance on this rule can oversimplify complex cognitive behavior.

Q9. What is cognitive offloading, and how do interfaces support it?**Model Answer:**

Cognitive offloading refers to using external tools or representations to reduce mental effort, such as notes, reminders, or visual indicators. Interfaces support cognitive offloading through calendars, to-do lists, progress indicators, and saved preferences. By shifting memory demands from the user to the system, designers reduce cognitive load and improve task performance. This is especially important for complex or long-term tasks.

4. Mental Models & Conceptual Understanding

Q10. What is a mental model, and why is it critical in HCI?**Model Answer:**

A mental model is the user's internal representation of how a system works, formed through experience, instruction, and interface cues. Users rely on mental models to predict system behavior and choose actions. When the interface aligns with the user's mental model, interaction feels intuitive. When it does not, users make systematic errors even if the system is technically correct.

Q11. Explain the thermostat example and its relevance to mental models.

Model Answer:

Many users treat thermostats like gas pedals, believing higher settings heat rooms faster, when in reality thermostats function as on/off control systems. This mismatch between the user's mental model and system behavior leads to incorrect usage. The example highlights the importance of designing interfaces that either match existing mental models or clearly teach new ones. Poor alignment results in persistent user confusion.

Q12. Describe Norman's Designer's Model, System Image, and User's Model.**Model Answer:**

The Designer's Model represents how designers believe the system works, while the System Image is what the system actually presents to users through interface, documentation, and feedback. The User's Model is the mental model users form based on the system image. Successful design ensures that the system image accurately communicates the designer's intent, enabling users to develop correct mental models.

5. Gulfs of Execution & Evaluation (VERY EXAMABLE)**Q13. What is the Gulf of Execution?****Model Answer:**

The Gulf of Execution is the gap between a user's goal and the actions required to achieve it using the system. It reflects how easily users can determine what actions are possible and how to perform them. Large gulfs occur when controls are hidden, unclear, or poorly labeled. Reducing this gulf requires visibility, signifiers, and intuitive mappings.

Q14. What is the Gulf of Evaluation?**Model Answer:**

The Gulf of Evaluation is the gap between the system's state and the user's ability to understand what happened after an action. If feedback is unclear or absent, users cannot determine whether their action was successful. This leads to uncertainty, repeated actions, or errors. Clear, immediate feedback bridges this gulf and supports user confidence.

Q15. How do visibility and feedback help bridge the gulfs?

Model Answer:

Visibility helps users understand what actions are possible, reducing the Gulf of Execution. Feedback helps users understand the results of their actions, reducing the Gulf of Evaluation. Together, they create a continuous loop of action and interpretation. Interfaces lacking either cause users to feel lost or out of control.

6. Direct Manipulation

Q16. Define direct manipulation and explain its core characteristics.**Model Answer:**

Direct manipulation is an interaction style where users interact directly with visible objects rather than using abstract commands. Its core characteristics are continuous representation of objects, physical actions instead of syntax, and rapid, reversible actions with immediate feedback. This style supports exploration and reduces fear of errors. It is especially effective for novice users.

Q17. Why are rapid reversible actions important in direct manipulation?**Model Answer:**

Rapid reversible actions allow users to undo mistakes instantly, reducing anxiety and encouraging exploration. This sense of safety promotes learning through trial and error. Without reversibility, users become cautious and interaction slows down. Undo and redo functions are essential components of user control.

7. Design Rules & Principles

Q18. What is the difference between principles, guidelines, and standards?**Model Answer:**

Principles are abstract, high-level truths that guide design thinking and require interpretation. Guidelines are recommended practices based on experience but are not mandatory. Standards are authoritative rules that must be followed, often defined by organizations or platforms. Together, they balance flexibility with consistency.

Q19. Explain the principle of visibility with an example.

Model Answer:

Visibility refers to making relevant controls and system status clearly perceivable to users. When users can see what actions are possible, they can act without guessing. Poor visibility forces users to rely on memory or trial and error. Examples include clearly labeled buttons and visible progress indicators.

Q20. Explain feedback and why it is essential in interaction design.**Model Answer:**

Feedback is the system's response to user actions, informing them of what has happened. It reassures users that the system is working and helps them interpret outcomes. Feedback can be visual, auditory, or tactile. Without feedback, users feel uncertain and may repeat actions unnecessarily.

Q21. What are constraints, and how do they prevent errors?**Model Answer:**

Constraints restrict the set of actions users can perform, preventing incorrect or dangerous operations. They can be physical, logical, cultural, or semantic. By guiding users toward valid actions, constraints reduce errors and decision complexity. Effective constraints feel helpful rather than restrictive.

Q22. Explain mapping and its role in logical design.**Model Answer:**

Mapping describes the relationship between controls and their effects in the system. Good mapping makes this relationship obvious and intuitive. Poor mapping forces users to think and remember arbitrary relationships. Logical spatial arrangement and clear labeling improve mapping quality.

Q23. Differentiate internal consistency and external consistency.**Model Answer:**

Internal consistency refers to uniform behavior within a single system, such as consistent button placement and interaction patterns. External consistency refers to alignment with conventions across different systems. Both reduce learning effort and errors. Violating either forces users to switch mental models.

Q24. Define affordance and perceived affordance.**Model Answer:**

Affordance refers to the properties of an object that suggest how it can be used. In interfaces, perceived affordances are learned visual cues that indicate possible actions. Because digital objects lack physical properties, designers rely heavily on conventions to communicate affordances. Weak affordances lead to user hesitation and confusion.

8. Accessibility & Inclusiveness

Q25. Define accessibility in interaction design.**Model Answer:**

Accessibility refers to designing products that can be used by people with disabilities, such as visual, auditory, motor, or cognitive impairments. It ensures that users do not require special adaptations to access core functionality. Accessibility is a usability requirement, not an optional feature. It improves design quality for all users.

Q26. Explain inclusiveness and how it differs from accessibility.**Model Answer:**

Inclusiveness aims to design for the widest possible range of users, regardless of disability, age, education, or socio-economic background. While accessibility focuses on specific impairments, inclusiveness considers diversity more broadly. Inclusive design benefits everyone by avoiding assumptions about “average” users. It promotes equity and participation.

Q27. Explain the three types of disability and their durations.**Model Answer:**

Disabilities can be sensory (vision or hearing), physical (movement or motor control), or cognitive (memory, learning, reasoning). Each type can be permanent, temporary, or situational. For example, a broken arm is temporary, while glare causing screen invisibility is situational. Recognizing this spectrum helps designers build flexible systems.

9. User Experience & Emotion

Q28. Define user experience (UX).

Model Answer:

User experience refers to a person's perceptions, emotions, and responses resulting from interaction with a system. It includes usability, aesthetics, emotional reactions, and meaning. UX goes beyond task completion to consider how interaction feels over time. A product can be usable but still deliver poor UX.

Q29. Explain pragmatic and hedonic qualities in UX.

Model Answer:

Pragmatic qualities relate to usefulness, efficiency, and goal achievement. Hedonic qualities relate to pleasure, stimulation, and personal meaning. Good UX balances both, ensuring tasks are easy while interaction is enjoyable. Neglecting either leads to functional but uninspiring products or fun but impractical ones.

Q30. Explain Norman's three levels of emotional design.

Model Answer:

Visceral design concerns immediate appearance and first impressions. Behavioral design focuses on usability and performance during interaction. Reflective design involves meaning, identity, and long-term satisfaction. Successful products address all three levels simultaneously.

10. Prototyping & Design Process

Q31. Why is prototyping essential in interaction design?

Model Answer:

Prototyping allows designers to explore ideas, test assumptions, and identify usability problems early. It reduces development risk by enabling rapid iteration before expensive implementation. Prototypes also help communicate design intent to stakeholders. Early feedback leads to better final products.

Q32. Explain low-fidelity prototyping and its advantages.

Model Answer:

Low-fidelity prototypes are rough, inexpensive representations such as sketches or paper mockups. They encourage experimentation and early feedback without emotional attachment. Users feel more comfortable criticizing them. Their simplicity keeps focus on structure and flow rather than aesthetics.

Q33. Explain high-fidelity prototyping and its disadvantages.**Model Answer:**

High-fidelity prototypes closely resemble the final product in look and interaction. They allow realistic usability testing and performance measurement. However, they require more time and resources. Designers may resist changes due to sunk cost effects.

11. Evaluation & Testing

Q34. What is usability testing, and what is its primary goal?**Model Answer:**

Usability testing involves observing representative users performing tasks to identify usability problems. Its goal is to improve the product, not to prove hypotheses. It focuses on effectiveness, efficiency, and satisfaction. Testing is iterative and informs design refinement.

Q35. Differentiate formative and summative evaluation.**Model Answer:**

Formative evaluation occurs early and aims to shape and improve design. Summative evaluation occurs later and assesses overall quality or compares alternatives. Formative tests are exploratory, while summative tests are judgment-oriented. Both are necessary at different stages.

Q36. What is triangulation in HCI research?**Model Answer:**

Triangulation involves using multiple methods, data sources, or evaluators to study the same phenomenon. It increases validity and reliability of findings. By combining qualitative and

quantitative approaches, designers gain deeper insights. It reduces bias inherent in single methods.

Q37. Explain ethnographic research and its outcomes.

Model Answer:

Ethnographic research involves observing users in their natural environments. It captures real practices, context, and implicit knowledge. Outputs often include personas, scenarios, and task models. This method is especially useful for early design stages.

Q38. What is participatory design?

Model Answer:

Participatory design actively involves users as collaborators in the design process. Users contribute ideas, feedback, and design decisions. This approach respects user expertise and increases acceptance. It shifts power from designers to shared ownership.

Q39. Explain card sorting and its purpose.

Model Answer:

Card sorting is a technique used to understand how users group and label information. Open card sorting allows users to define categories, while closed sorting tests predefined structures. It informs information architecture and navigation design. It reduces mismatches between designer and user organization.

Q40. What is technology myopia, and why should designers avoid it?

Model Answer:

Technology myopia is the tendency to focus excessively on technical solutions while ignoring human and social context. It leads to systems that work technically but fail in real use. Designers must consider workflows, emotions, culture, and communication. Avoiding myopia ensures systems fit real human lives.